# Build the safety boundary before the AI

From written requirements to a validated, API-backed presentation pipeline.

## 01 / ARCHITECTURE JOURNEY

| | |
|---|---|
| **7** | task milestones |
| **4** | validation gates |
| **9** | approved components |
| **0** | model calls in this scope |

### I / FOUNDATION & BOUNDARY
Define what is allowed

**T1 / REQUIREMENTS**

#### Domain & specifications

BR / FR / VR traceability; runtime boundaries; seven non-negotiable rules.

**T2 / CONTRACT**

#### UISpec v1 + JSON Schema

specVersion 1.0; closed vocabulary, props, actions and presentation policy.

**T3 / VALIDATOR**

#### Deterministic acceptance

Four ordered gates report all rejections with stable code, field path and gate.

### II / UI EXECUTION
Render only approved input

**T4 / CONTROLLED RENDERER**

#### A closed component registry

Nine components; exhaustive TypeScript map; own-property lookup. ApprovedUISpec records provenance from the offline Python validation gate.

**T5 / RUNNABLE UI**

#### Prove the browser path

Vite + React + strict TypeScript. Static approved fixtures render through the registry before a network boundary is introduced.

### III / RUNTIME & SEAM
Serve only validated output

**T6 / DOMAIN RUNTIME**

#### Move trust to the server

FastAPI validates each fixture against complete server-owned GroundedContext on every request. The client accepts a checked response envelope.

**T6A / CONTEXT SEAM**

#### Type the presentation request

Required page, view and locale become an immutable ContextPayload. The service checks page / locale against its owned context.

**DESIGN LESSON** **Contract → validator → renderer → browser → API → typed context seam.**

# One request. One deterministic trust boundary.

The caller selects a presentation. The Domain Runtime supplies the grounded context.

## 02 / AS-BUILT REQUEST & RESPONSE

**UI RUNTIME / :5173**

### React browser

Fixed API path and method.
Same-origin Vite development proxy for /api and /health.

**POST /api/v1/ui/landing-page**

```
{
"page": "landing",
"view": "overview",
"locale": "en-US"
}
```

**DOMAIN RUNTIME / FastAPI · 127.0.0.1:8000**

**1 / TRANSPORT**

### Validate the body

LandingPageRequest: all fields required; extra="forbid". Bad inputs → HTTP 400.

**2 / DOMAIN VALUE**

### ContextPayload

Frozen, slotted enum value. Presentation request only; no metrics or contextId.

**3 / ORCHESTRATION**

### LandingService

Check page / locale against server context. Never rewrite a fixture to fit the caller.

**4 / SERVER-OWNED INPUT**

### Fixture + GroundedContext

Approved presentation fixture and complete grounded context; supplied by the server.

**5 / ACCEPTANCE GATE**

### Deterministic Python validator

Structure → vocabulary → grounding → text policy. The same input yields the same result.

**6 / HTTP 200**

### Accepted response

{spec, bindings, meta}; meta carries contextId, source: "fixture", validation: "VALID".

**7 / CLIENT ACCEPTANCE**

### unknown → checked envelope

acceptLandingResponse() narrows the response before registry-only rendering.

**REJECTION / FAIL CLOSED**

### Sanitized error state

Invalid fixture → 500 VALIDATION_FAILED. Codes stay in logs; the invalid spec is never returned.

GET /health → {"status":"ok"} indicates process liveness only. It does not establish inspection readiness.

# Schema-valid is only the first gate

Four ordered validator gates • 13 stable rejection codes • all rejections reported.

03 / DEFENCE IN DEPTH

**01** STRUCTURE

## Well formed and bounded

Reject malformed structure and size / depth abuse.

**E_SCHEMA · E_SIZE_LIMIT · E_DEPTH_LIMIT**

**02** VOCABULARY

## Only approved building blocks

Reject unknown components and invalid component props.

**E_UNKNOWN_COMPONENT · E_PROP_INVALID**

**03** GROUNDING

## Every claim traces to context

Reject invented metrics, categories, actions or capabilities.

**E_UNGROUNDED_METRIC · E_UNKNOWN_DEFECT_CATEGORY**
**E_UNKNOWN_ACTION · E_UNGROUNDED_CAPABILITY**
**E_CONTEXT_MISMATCH**

**04** TEXT POLICY

## Presentation never becomes code or verdict

Reject disallowed markup, URLs, code-like text and decision language.

**E_TEXT_POLICY · E_CODE_LIKE_CONTENT**
**E_DECISION_LANGUAGE**

THE ONLY RENDER PATH

## Closed registry · 9 components

- Hero
- Heading
- Text
- FeatureCard
- FeatureGrid
- CTA
- Alert
- InfoPanel
- ProgressIndicator

COMPILE + BUILD CONTROLS

## Provenance, then enforcement

ApprovedUISpec brand; exhaustive registry map; own-property lookup; safety lint.

MEANING OF VALID

## Presentation validity only

VALID never means that a board passed inspection.

# Keep the implemented contract explicit

Future extension belongs behind the service and ahead of the validator.

04 / CONTRACT, GOVERNANCE & NEXT SEAM

AS BUILT / REQUIRED REQUEST FIELDS

| | |
|---|---|
| **page** | landing |
| **view** | overview \| minimal |
| **locale** | en-US |

No defaults. Additional fields → HTTP 400. Query parameters do not select the presentation.

SEVEN ARCHITECTURAL RULES

01 UI specification only; never executable content.
02 Registry is the only render path.
03 Validate; bounded repair / fallback are future wiring.
04 Model reads no data directly.
05 Model makes no quality decisions.
06 Model invents nothing; claims must be grounded.
07 No authentication subsystem in this phase.

REPORTED VERIFICATION / 06 OCT 2026

## 217 pytest passed · 88 UI tests passed

Typecheck, safety lint and build clean; source report, not a fresh run.

FUTURE / NOT IMPLEMENTED IN TASKS 1–6A

## A model gateway plugs into the seam

Service → future model-generated UISpec → unchanged validator → controlled renderer. One bounded repair and deterministic fallback are specified, but not wired. No model calls, database, authentication subsystem, camera / PLC / MES integrations or production deployment.

BUSINESS DECISIONS / CANDIDATES ONLY

## Line, board type and inspection mode

Value sets remain undecided. Manufacturing owns line IDs; Manufacturing / Quality own board taxonomy; Quality owns modes.

CONTRACT CHANGE / GOVERNED TOGETHER

## No generated_for response field

Request echoing is not adopted. New dimensions require coordinated contract, schema, deterministic test and eval changes.

SOURCE AUTHORITY

CLAUDE.md → specs/spec.md → model-contracts.md → ui-spec.schema.json.
Repository wins if this summary differs.